Price efficiency through variance decomposition in ESG narrative

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Academic Year: 2025–2026

In order to obtain the joint degree:

Master 120 en Sciences économiques, Orientation générale,
Finalité approfondie (UCL/UNamur)

and

Dottore Magistrale in Finance and Economics (UNIMI)

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1 Introduction

This paper asks whether the rise of ESG investing has changed the composition of European equity-return variance, measured as the balance between firm-specific information, market-wide information and transitory noise. The question sits at the intersection of asset-pricing research on price informativeness and the recent ESG literature: if ESG-relevant information is increasingly priced at the firm level, one should observe changes in how return variance is allocated across these components.

To answer this question a formal notion of what "information" and "noise" mean in stock returns is required. Brogaard et al. (2022) provide a unified return variance decomposition that explicitly distinguishes noise from information and, within information, separates market-wide from firm-specific shocks, further partitioning the latter into a private component (revealed through informed trading) and a public component (revealed through company announcements and news). The starting point is the classical permanent–transitory decomposition of Beveridge and Nelson (1981), in which the log price $p_{i,t}$ of a security is written as the sum of an efficient (random-walk) component $m_{i,t}$ and a transitory pricing error $s_{i,t}$:

$$p_{i,t} = m_{i,t} + s_{i,t}, \qquad m_{i,t} = m_{i,t-1} + \mu + w_{i,t}, \qquad \mathbb{E}_{t-1}[w_{i,t}] = 0, \tag{1.1}$$

where μ is the discount-rate drift and $w_{i,t}$ are the unpredictable innovations to fundamental value. The return $r_{i,t} = p_{i,t} - p_{i,t-1}$ therefore is divided in: a) the permanent information shock $w_{i,t}$ and b) the change $\Delta s_{i,t}$ in the pricing error, which is by construction transitory and captures over- or under-reaction, illiquidity, price impact and microstructure frictions.

Brogaard et al. further partition the information innovation $w_{i,t}$ along an empirically meaningful three-way split:

$$w_{i,t} \;=\; \underbrace{\theta_{rm}\,\varepsilon_{rm,t}}_{\text{Market-wide information}} \;+\; \underbrace{\theta_x\,\varepsilon_{x,i,t}}_{\text{Private firm-specific info}} \;+\; \underbrace{\theta_r\,\varepsilon_{r,i,t}}_{\text{Public firm-specific info}}, \tag{1.2}$$

where $\varepsilon_{rm,t}$ is the structural shock to the market return, $\varepsilon_{x,i,t}$ the structural shock to a measure of signed dollar volume (used as a microstructure proxy for informed trading), and $\varepsilon_{r,i,t}$ the residual shock to the stock return after the first two have been partialled out. The scalars $\theta_{rm}, \theta_x, \theta_r$ are the long-run cumulative responses of $r_{i,t}$ to unit innovations of each type, and they capture the *permanent* (informational) content of each shock under the Beveridge–Nelson interpretation. As detailed in Section 3.2.2, the baseline design of this work aggregates the private and public pieces into a single firm-specific information share, so that the object actually estimated matches the three-way split (market-wide, firm-specific, noise) used to state the research question above.

This formal separation is also what distinguishes the approach from the long-standing synchronicity literature, where traditional measures based on one minus market-model R^2 cannot separate firm-specific information from firm-specific noise (Jin and Myers, 2006; Morck et al., 2000; Roll, 1988). The decomposition in equations (1.1)–(1.2) is useful precisely because it separates information from noise at the firm-period level, allowing to test the ESG channel on the relevant object instead of on a reduced-form proxy.

The contribution is mainly a direct test of whether within-firm changes in ESG scores are associated with shifts in the information/noise mix of returns in a large European panel. Methodologically, it adapts the decomposition framework of Brogaard et al. (2022) from annual U.S. applications to firm-quarter estimation in Europe, making the design compatible with quarterly ESG and control variables while increasing the frequency of the estimated object of interest.

The empirical motivation comes from two strands of recent work. First, equilibrium models of sustainable investing imply that changes in investors' ESG preferences can affect prices, expected returns and volatility, potentially altering the relative importance of firm-specific and market-wide shocks (Pástor et al., 2022). Second, work on disclosure and measurement emphasizes that ESG signals can be noisy and heterogeneous across providers, which can attenuate or reshape the mapping from sustainability information to prices (Berg et al., 2022; Christensen et al., 2021). In parallel, narrative-attention evidence shows that text-based information environments can move price informativeness even when signals are imperfect (Dim et al., 2023; Hassan et al., 2019).

These strands imply a testable prediction but not a unique sign. If ESG-score revisions signal the arrival of relevant firm information, firm-specific information shares should rise and, if uncertainty is reduced, noise should fall. If instead ESG repricing is driven mainly by broad demand shifts or by heterogeneous interpretations of ESG metrics, market-wide shares may rise and noise need not decline. Existing evidence has mostly studied average returns and volatility rather than the structural composition of return variance, leaving this question open.

The empirical answer in this dataset is clear: the baseline two-way fixed-effects estimates do not detect statistically significant effects of quarterly ESG-score changes on firm-specific, market-wide or noise shares. In this sense, the data do not confirm a strong version of the prediction that ESG-score revisions systematically reshape variance composition over 2015–2025 in the STOXX Europe 600 sample. This null is informative but conditional on measurement limits discussed later, especially the low-frequency updating and persistence of rating-based ESG measures.

The remainder of the paper is organised as follows. Section 2 reviews the related literature. Section 3 presents the data, the variance decomposition framework, and the panel-regression specification. Section 4 reports and discusses the empirical results. Section 5 concludes.

2 Literature Review

The central identification issue in this paper is the one at the core of the synchronicity literature: whether idiosyncratic return variation reflects firm-specific information or firm-specific noise. The classical decomposition of stock returns into cash-flow and discount-rate news (Campbell, 1991; Campbell and Shiller, 1988; Chen et al., 2013; Chen and Zhao, 2009) and the permanent–transitory logic of Beveridge and Nelson (1981) provide the conceptual background, but the key empirical challenge is to isolate information from transitory pricing-error variation in observed returns.

The intuition behind the synchronicity literature is made explicit here by a model that is a compact version of Jin and Myers (2006), augmented with noise in the spirit of Black (1986); it is the same theoretical device from which Brogaard et al. (2022) derive their four variance shares, and reproducing it here makes transparent both what the market-model R^2 measures and what it cannot.

Let the cash-flow news accruing to firm i in period t be the sum of a systematic and an idiosyncratic component,

$$\delta_{i,t} \;=\; \beta_i f_t \;+\; z_{i,t}, \qquad f_t \sim (0, \sigma_f^2), \quad z_{i,t} \sim (0, \sigma_{z,i}^2), \tag{2.1}$$

where $\delta_{i,t}$ denotes total cash-flow news for firm i in period t (the unexpected information about future firm cash flows coming from both economy-wide developments and firm-specific shocks), with $f_t \perp z_{i,t}$, $z_{i,t} \perp z_{j,t}$ for $i \neq j$, and β_i the firm's loading on the common factor. Not all of this news reaches prices within the period. Following the opacity logic of Jin and Myers (2006), in which insiders absorb part of the firm-specific cash flow and withhold it from outside investors, let $\kappa_m \in [0,1]$ and $\kappa_i \in [0,1]$ denote the fractions of systematic and firm-specific news that are capitalised into the efficient price in period t. The efficient price component of equation (1.1) then evolves as $m_{i,t} = m_{i,t-1} + \mu + w_{i,t}$ with

$$w_{i,t} \;=\; \underbrace{\kappa_m \beta_i f_t}_{\text{market-wide}} \;+\; \underbrace{\lambda_i \kappa_i z_{i,t}}_{\text{private firm-specific}} \;+\; \underbrace{(1-\lambda_i)\kappa_i z_{i,t}}_{\text{public firm-specific}}\,, \tag{2.2}$$

where $\lambda_i \in [0,1]$ is the fraction of firm-specific news revealed through informed trading rather than through disclosure. Equation (2.2) is exactly the partition in equation (1.2): the structural loadings and shocks map into the primitives as $\theta_{rm}\varepsilon_{rm,t} = \kappa_m \beta_i f_t$, $\theta_x \varepsilon_{x,i,t} = \lambda_i \kappa_i z_{i,t}$ and $\theta_r \varepsilon_{r,i,t} = (1-\lambda_i)\kappa_i z_{i,t}$. The aggregation performed in the first deviation of Section 3.2.2 amounts to leaving λ_i unidentified while still recovering $\kappa_i z_{i,t}$ in full.

Observed prices differ from $m_{i,t}$ by a transitory pricing error, which is assumed to be mean-reverting,

$$s_{i,t} \;=\; \rho_i\, s_{i,t-1} + u_{i,t}, \qquad |\rho_i| < 1, \qquad u_{i,t} \sim (0, \sigma_{u,i}^2), \tag{2.3}$$

with $u_{i,t}$ orthogonal to f_t and $z_{i,t}$. The parameter $\sigma_{u,i}^2$ governs the magnitude of over- and under-reaction, price pressure and microstructure frictions, while ρ_i governs the speed at which they revert. Realised returns are $r_{i,t} = w_{i,t} + \Delta s_{i,t}$, and since the three building blocks are mutually orthogonal,

$$\mathrm{Var}(r_{i,t}) \;=\; \underbrace{\kappa_m^2 \beta_i^2 \sigma_f^2}_{\text{MktInfo}_i} \;+\; \underbrace{\kappa_i^2 \sigma_{z,i}^2}_{\text{FirmInfo}_i} \;+\; \underbrace{\frac{2\sigma_{u,i}^2}{1+\rho_i}}_{\text{Noise}_i}\,. \tag{2.4}$$

Dividing each term by the total delivers the three shares of Section 3.1.2, and equation (2.4) is the theoretical counterpart of the empirical decomposition (3.3).[1]

[1] In the model the sum of the components coincides with total return variance, because information and noise are independent by construction. The empirical implementation relaxes that assumption and therefore normalises by the sum of the components rather than by total variance, so that the shares continue to add to one; Brogaard et al. (2022) show that the neglected covariance term has a negligible effect on the estimates.

The synchronicity literature does not estimate (2.4). It runs the static market-model projection

$$r_{i,t} \;=\; a_i \;+\; b_i\, r_{m,t} \;+\; e_{i,t}, \qquad R_i^2 \;=\; \frac{b_i^2\,\mathrm{Var}(r_{m,t})}{\mathrm{Var}(r_{i,t})}, \tag{2.5}$$

and interprets $1 - R_i^2$ as the amount of firm-specific information in prices (Jin and Myers, 2006; Morck et al., 2000; Roll, 1988). In a well-diversified index the idiosyncratic and noise components wash out, so that $r_{m,t} = \kappa_m f_t$ and $b_i = \beta_i$. Substituting into (2.5) and using (2.4) yields the central mapping between the two literatures:

$$R_i^2 \;=\; \mathrm{MktInfoShare}_i, \qquad 1 - R_i^2 \;=\; \mathrm{FirmInfoShare}_i \;+\; \mathrm{NoiseShare}_i. \tag{2.6}$$

The synchronicity measure of Morck et al. (2000), $\Psi_i = \log\!\Big(R_i^2/(1 - R_i^2)\Big)$, is therefore a monotone transformation of the ratio of the market-wide share to the sum of the other two.[2]

Equation (2.6) makes the ambiguity precise. Differentiating,

$$\frac{\partial(1 - R_i^2)}{\partial \kappa_i} \;>\; 0, \qquad \frac{\partial(1 - R_i^2)}{\partial \sigma_{u,i}^2} \;>\; 0, \tag{2.7}$$

so a fall in synchronicity is equally consistent with prices impounding more firm-specific information ($\kappa_i \uparrow$) and with prices becoming noisier ($\sigma_{u,i}^2 \uparrow$). Worse, the two channels can offset one another exactly: any joint movement leaving $\kappa_i^2\sigma_{z,i}^2 + 2\sigma_{u,i}^2/(1+\rho_i)$ unchanged is invisible to R^2. The measure is a valid proxy for firm-specific information only under the auxiliary restriction that the noise share is constant across firms and over time, a restriction that the cross-sectional evidence rejects: Brogaard et al. (2022) document, in the main U.S. sample, a noise share that falls monotonically from 35.56% in the smallest size quartile to 16.92% in the largest (their Table 2, Panel C). The same logic produces the sign reversal they emphasise, whereby an improvement in efficiency through lower idiosyncratic noise mechanically raises R^2 and would be read, under the traditional interpretation, as a deterioration.

The model also disciplines the hypotheses tested below, since each of the mechanisms discussed in the introduction acts on a different primitive of equation (2.4). If ESG-score revisions convey value-relevant firm-specific news, they raise κ_i, increasing FirmInfoShare and lowering the other two shares. If instead ESG repricing is driven by broad, correlated shifts in investor tastes of the kind modelled by Pástor et al. (2022), the relevant shock enters the systematic block and raises MktInfoShare. If the divergence in ESG ratings documented by Berg et al. (2022) generates disagreement rather than information, it raises $\sigma_{u,i}^2$ and hence NoiseShare. Under $1 - R^2$ the first and third mechanisms are observationally equivalent, by (2.7); under the decomposition they carry opposite signs on distinct dependent variables.

This is where Brogaard et al. (2022) provide the relevant new framework, building on structural-VAR tools (Lütkepohl, 2005; Sims, 1980) and long-run identification logic (Blanchard and Quah, 1989). The older synchronicity interpretation, developed by Roll

[2] The identity in (2.6) is exact only in the static model above. Empirically, R^2 comes from a contemporaneous projection whereas the variance shares come from long-run cumulative responses in the VAR of Section 3.2, so the two objects are closely related but not numerically identical.

(1988), Morck et al. (2000) and Jin and Myers (2006), linked low market-model R^2 to richer firm-specific information in prices, an influential but, as shown above, not point-identified reading of the residual.

Subsequent evidence reinforces this caution. Griffin et al. (2010) show that alternative efficiency metrics can deliver conflicting rankings across markets, while Wurgler (2000) and Bond et al. (2012) emphasize that the real effects of prices depend on the mechanism through which information is aggregated and transmitted. Using a different approach, Bai et al. (2016) and Farboodi et al. (2022) document changes in price informativeness over time that do not map one-to-one into the synchronicity interpretation of R^2.

Recent work further suggests that information environments are shaped by narratives and attention as well as by fundamentals, and that these channels can affect price informativeness in ways not captured by market-model residual variation (Dim et al., 2023; Hassan et al., 2019). Taken together, the literature points to a practical conclusion: if the objective is to study how return variation is allocated across informational and non-informational components, variance decomposition is the appropriate empirical tool.

3 Methodology

The empirical strategy of this work proceeds in two consecutive steps. First, daily return data are summarised at the firm–quarter level through a variance decomposition based on the framework of Brogaard et al. (2022). Second, the resulting quarter-by-quarter variance shares are stacked into a firm-quarter panel and related to a set of firm-level controls through a fixed-effects regression. This section documents the data underlying both stages (Section 3.1), the original Brogaard et al. (2022) variance decomposition and differences with the current work (Section 3.2), and the panel-regression specification (Section 3.3).[3]

3.1 Data

3.1.1 Chosen Sample and Sources

All the stock tickers are obtained from Bloomberg through the `EQS` (Equity Screening) function and the standard time-series download. They are subsequently used as input in the Bloomberg Excel extension to create datasheets for both daily and quarterly variables for each stock.

The starting universe is the STOXX Europe 600 Price Index in euro, with the EQS filter restricting to active tickers with current market capitalisation above $1 billion at extraction time, a floor chosen to keep ESG-rating coverage adequate and daily prices liquid enough for the firm-quarter variance decomposition of Section 3.2.[4] This returns 596 individual stocks plus the index. Because the screen is applied at extraction time,

[3]The code for the full pipeline, from data processing through the decomposition to the panel regression, is available at https://github.com/FrancescoDotti/Price-efficiency-through-variancedecomposition-in-ESG-narrative.

[4]Exact filter: *Indices* = "STOXX Europe 600 Price Index EUR"; *Trading Status* = "Active"; *Current Market Capitalisation* > $1 billion.

survivorship bias is present at the constituent level; the firm fixed effects in the panel block its first-order implications. The window runs from January 2015 to December 2025 (44 calendar quarters, 2,870 trading days). The start date and reference index have been chosen because in the EU context there has always been a strong interest in regulating and pushing ESG products with a decisive acceleration after the EU Action Plan of 2018 and the entry into force of SFDR/Taxonomy/MiFID II between 2021 and 2023.

Tables 3.1 and 3.2 report the geographic and sectoral breakdown of the cross-section. The seven largest national groups account for $\sim$ 80% of the panel while financial firms account for roughly 20%, with industrials, healthcare and consumer non-cyclicals completing the largest sectors.

Table 3.1: Sample composition by listing venue

Venue	N	Venue	N	Venue	N
United Kingdom (LN)	130	Sweden (SS)	56	Belgium (BB)	19
France (FP)	71	Italy (IM)	40	Finland (FH)	19
Germany (GR)	69	Netherlands (NA)	34	Poland (PW)	12
Switzerland (SW)	59	Denmark (DC)	25	Austria (AV)	8
		Spain (SM)	25	Ireland (ID)	6
		Norway (NO)	19	Portugal (PL)	4
Total: 596 stocks					

Notes: Stocks are grouped by their Bloomberg exchange-code suffix.

Table 3.2: Sample composition: ten largest GICS industries

Industry	N	Industry	N
Banks	46	Pharmaceuticals	19
Machinery	37	Oil, Gas & Consumable Fuels	17
Capital Markets	30	Food Products	16
Insurance	30	Financial Services	16
Chemicals	20	Aerospace & Defense	15
Remaining: 350 stocks across ~ 50 *GICS labels*			

Daily prices and returns. The decomposition uses closing prices (`PX_LAST` on Bloomberg), adjusted for corporate actions but not for cash dividends (which has negligible implications for the noise share and no correlation with the ESG score), and works throughout in logs: $p_{i,t}$ denotes the log of the closing price, and returns are the corresponding log returns,

$$r_{i,t} = p_{i,t} - p_{i,t-1}, \tag{3.1}$$

exactly as in the log-price decomposition of Section 1 and in Brogaard et al. (2022). Closing prices are forward-filled for at most five consecutive days.

Quarterly control and ESG variables. Firm-level fundamentals enter at quarterly frequency and Table 3.3 reports the share of non-missing firm-quarters for each variable used or available to the regression.

Table 3.3: Quarterly Bloomberg variables: coverage in the firm-quarter panel

Bloomberg field	Coverage	Role in the regression
ESG block		
`ESG_SCORE`	88.1%	main regressor (in first differences)
`GOVERNANCE_SCORE`	89.9%	robustness
`ENVIRONMENTAL_SCORE`	88.1%	robustness
`SOCIAL_SCORE`	88.1%	robustness
Size and ownership		
`CUR_MKT_CAP`	99.8%	control (in log)
`PCT_INSIDER_SHARES_OUT`	98.3%	robustness
Financial-statement items		
`BS_TOT_ASSET`	64.9%	robustness
`HEADLINE_BVPS`	64.0%	robustness
`PROF_MARGIN`	62.4%	robustness
`ASSET_TURNOVER`	60.0%	robustness
`FNCL_LVRG`	57.9%	robustness
`RETURN_COM_EQY`	57.2%	robustness
`CASH_FLOW_PER_SH`	53.4%	robustness
`HEADLINE_CAPEX`	50.9%	robustness

Notes: Coverage is the share of non-missing firm-quarters computed over the 24,810-cell decomposition sample (the firm-quarters that enter the analysis), not over the raw 596×44 grid.

The regressor of interest is the Bloomberg `ESG_SCORE`, a composite ESG rating computed against an industry-specific peer group. This variable is approximately constant within firm-quarters, so ΔESG isolates the news content of a score revision. The distribution is sharply peaked at zero (median quarterly change of 0, 25th and 75th percentiles both 0, since scores typically update once or twice a year) with a small positive mean of 0.05.

The financial-statement items have coverage between 51–65% because European firms file at half-year or annual cadence in roughly half the sample. They are reserved for robustness specifications: including any of them in the baseline shrinks the estimation sample by $\sim 40\%$ in a way that is correlated with reporting frequency, which would amplify a reporting-frequency selection that the firm fixed effects already absorb.

Derived variables. There are variables like market capitalisation which undergo a logarithmic transformation, $\log\Big(\max\{\texttt{CUR\_MKT\_CAP}_{i,t},\, 1\}\Big)$, which corrects for the strong right-skew of the raw series; the $\max\{\cdot, 1\}$ floor handles a handful of near-zero corporate-event cells. The second is a quarterly realised volatility measure built directly from the daily log returns of equation (3.1),

$$\widehat{\sigma}_{i,t} \;=\; \mathrm{std}\Big(r_{i,d}\Big)_{d \in t} \;\times\; \sqrt{65}, \tag{3.2}$$

where $d \in t$ indexes the trading days that fall inside quarter t and $\sqrt{65}$ scales the daily standard deviation to the quarterly horizon under a convention of 65 trading days per quarter.

3.1.2 Variance Decomposition Shares.

The dependent variables of the regression are firm-quarter variance shares produced by the variance decomposition rather than extracted from Bloomberg. They are constrained by construction to lie in $[0, 100]$ and to sum to 100% within each cell:

- $\text{MktInfoShare}_{i,t}$, the share attributable to market-wide information shocks;
- $\text{FirmInfoShare}_{i,t}$, the share attributable to firm-specific information shocks (see the first deviation below);
- $\text{NoiseShare}_{i,t}$, the share attributable to transitory pricing-error variation.

Table 3.4 reports pooled summary statistics across the 24,810 firm-quarter cells that survive the decomposition filters. Firm-specific information dominates the cross-section (66.8% on average); market-wide information accounts for 26.7% with substantial dispersion ($\sigma = 18.9$ pp); noise accounts for the residual 6.5% ($\sigma = 6.2$ pp). The firm-specific share is somewhat higher than the 61% reported by Brogaard et al. (2022) for the U.S. sample 1960–2015, consistent with the European market being more fragmented (as market-info share is a measure of the tendency of firms to follow a broader index) and with the pooling of public and private firm-specific information into a single share, discussed below.

Table 3.4: Variance-decomposition outputs: pooled summary statistics

Variable	N	Mean	Std. Dev.	p25	Median	p75
`MktInfoShare` (%)	24,810	26.70	18.90	10.76	24.27	39.96
`FirmInfoShare` (%)	24,810	66.77	19.35	53.15	69.19	82.50
`NoiseShare` (%)	24,810	6.53	6.18	2.03	4.70	9.20

Notes: Sample: 24,810 firm-quarter observations between 2015-Q1 and 2025-Q4, 595 distinct stocks. The three shares sum to 100% within each firm-quarter cell.

Figure 3.1 plots the cross-sectional average of the three shares by year. The market-wide share spikes in 2020 (the covid-19 shock) and again, more mildly, around 2022 (the energy-price shock), consistent with both being euro-area-wide repricing episodes; the firm-specific share moves inversely, as the adding-up constraint of Section 4.1 requires; and the noise share ranges between 5% and 23% across the sample, with its highest values concentrated in 2016–2017.

3.2 The Brogaard et al. variance decomposition

3.2.1 Conceptual framework

Brogaard et al. (2022) provide a unified return variance decomposition that explicitly distinguishes *noise* from *information* and, within information, separates market-wide from firm-specific shocks, further partitioning the latter into a private component (revealed through informed trading) and a public component (revealed through company announcements and news). Section 1 lays out the conceptual backbone of this

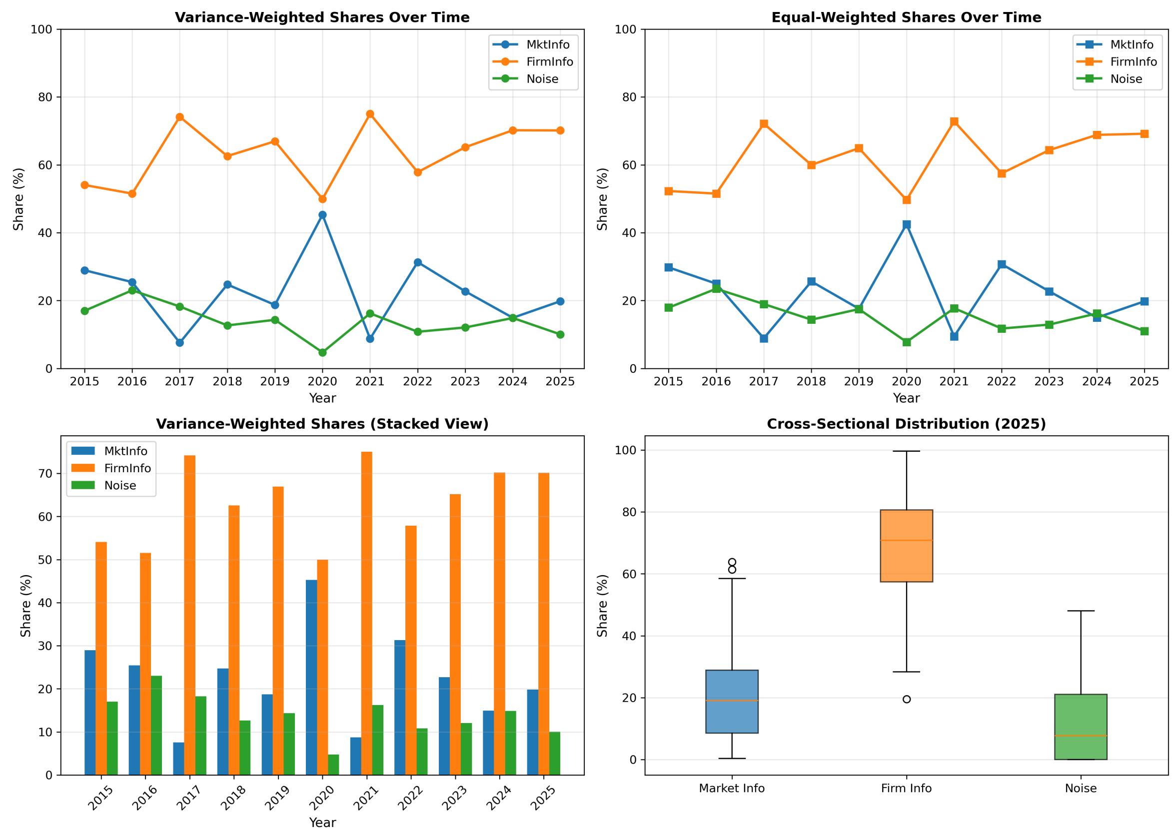

Figure 3.1: Variance-decomposition shares over time, by year

Notes: Cross-sectional average of `MktInfoShare`, `FirmInfoShare` and `NoiseShare` (%) for each calendar year between 2015 and 2025, variance-weighted (left) and equal-weighted (right). This figure aggregates to annual frequency for visual clarity; it is produced from a separate annual run of the same decomposition methodology described in Section 3.2, not from an averaging of the quarterly shares used in the regression panel.

decomposition. The associated variance components and shares follow by construction:

$$
\begin{aligned}
\text{MktInfo}_{i,t} &= \theta_{rm}^2\,\sigma_{\varepsilon_{rm}}^2, \\
\text{PrivateInfo}_{i,t} &= \theta_x^2\,\sigma_{\varepsilon_x}^2, \\
\text{PublicInfo}_{i,t} &= \theta_r^2\,\sigma_{\varepsilon_r}^2, \\
\text{Noise}_{i,t} &= \sigma_s^2 \;=\; \text{Var}\Big(\Delta s_{i,t}\Big),
\end{aligned}
\tag{3.3}
$$

with shares obtained by normalising each component by the total of the four. Because the structural shocks are contemporaneously uncorrelated by construction, the variance contributions are additive.

The Brogaard et al. (2022) approach then builds a three-variable structural VAR with five lags and contemporaneous restrictions ordered as $(r_{m,t}, x_{i,t}, r_{i,t})$:

$$
\begin{aligned}
r_{m,t} &= \textstyle\sum_{l=1}^{5} a_{1,l} r_{m,t-l} + \sum_{l=1}^{5} a_{2,l} x_{i,t-l} + \sum_{l=1}^{5} a_{3,l} r_{i,t-l} + \varepsilon_{rm,t}, \\
x_{i,t} &= b_{1,0} r_{m,t} + \textstyle\sum_{l=1}^{5} b_{1,l} r_{m,t-l} + \sum_{l=1}^{5} b_{2,l} x_{i,t-l} + \sum_{l=1}^{5} b_{3,l} r_{i,t-l} + \varepsilon_{x,i,t}, \\
r_{i,t} &= c_{1,0} r_{m,t} + c_{2,0} x_{i,t} + \textstyle\sum_{l=1}^{5} c_{1,l} r_{m,t-l} + \sum_{l=1}^{5} c_{2,l} x_{i,t-l} + \sum_{l=1}^{5} c_{3,l} r_{i,t-l} + \varepsilon_{r,i,t}.
\end{aligned}
\tag{3.4}
$$

The ordering encodes three identification restrictions: (i) the market is exogenous to the individual stock within the same day, since each stock is a negligible component of the index; (ii) trading in stock i can be contemporaneously caused by market moves but cannot itself move the market contemporaneously; and (iii) within a given day, stock returns can react to both market and trading shocks contemporaneously, while public information remains the residual.

The empirical strategy here is to estimate the reduced-form version of equation (3.4) stock-by-stock and year-by-year, recover the structural innovations through successive regressions of the reduced-form residuals, feed each unit structural shock through the reduced-form impulse-response function out to $t = 15$ days (three times the lag length), and combine the resulting θ's with the estimated σ_ε^2 to obtain the four variance components in equation (3.3). Variance shares are normalised to sum to one within each (stock, year).

3.2.2 Deviations from baseline model

The decomposition used here departs from Brogaard et al. (2022) in two main ways.

The first drops the volume regressor $x_{i,t}$ altogether, reducing the structural VAR to two equations in $(r_{m,t}, r_{i,t})$:

$$
\begin{aligned}
r_{m,t} &= a_0 + \textstyle\sum_{l=1}^{p} a_{1,l} r_{m,t-l} + \sum_{l=1}^{p} a_{2,l} r_{i,t-l} + e_{rm,t}, \\
r_{i,t} &= c_0 + \textstyle\sum_{l=1}^{p} c_{1,l} r_{m,t-l} + \sum_{l=1}^{p} c_{2,l} r_{i,t-l} + e_{r,t},
\end{aligned}
\tag{3.5}
$$

estimated as a reduced-form VAR with an intercept. Identification of the contemporaneous block is reduced to a single restriction: stock returns can react to the market return within the same day, while the market is exogenous to the individual stock. The contemporaneous loading coefficient ϕ linking the two reduced-form innovations is recovered from a univariate projection,

$$
\phi \;=\; \frac{\text{Cov}(e_{rm,t},\, e_{r,t})}{\text{Var}(e_{rm,t})},
\tag{3.6}
$$

which corresponds to the OLS slope of $e_{r,t}$ on $e_{rm,t}$ (with zero intercept by construction, since reduced-form residuals are mean-zero). The structural variances are then $\sigma^2_{\varepsilon rm} = \sigma^2_{e_{rm}}$ and $\sigma^2_{\varepsilon_r} = \sigma^2_{e_r} - \phi^2\, \sigma^2_{e_{rm}}$.

The main reason behind this choice is that identifying private firm-specific information relies on assumptions about market microstructure that are quite strong, especially given the low relevance of that extra split for answering the research question here. The resulting aggregated firm-specific component turns out to be almost identical to the sum of the private and public firm-specific components.

Table 3.5 shows this. Running the full three-variable decomposition (which separates private from public firm-specific information) and the two-variable decomposition used here on the same daily panel, the aggregated firm-specific share from the latter tracks the sum of the private and public shares from the former almost exactly: across the 6,232 stock-year cells the two series have a cross-sectional correlation of 0.99, average shares of 62.3% and 62.6%, and a mean absolute difference of roughly 1.8 percentage points (never above 2.3 in any year). Dropping the volume equation therefore costs essentially nothing in terms of the total firm-specific information recovered, while shedding the strong microstructure assumptions needed to split it. The same near-identity carries over to the quarterly windows actually used in the paper: re-estimating both decompositions at quarterly frequency leaves the cross-sectional correlation at 0.90, and the two pooled averages differ by no more than 2.5 percentage points, even though the typical individual stock-quarter cell differs by more (mean absolute difference $\approx$ 6.9 pp, as noted in Table 3.5).

Table 3.5: Aggregated firm-specific share vs. private + public, by year

Year	N	Aggregated (2-var, %)	Private + Public (3-var, %)	Corr.
2015	533	52.3	52.5	0.987
2016	540	51.8	52.2	0.990
2017	549	72.1	72.0	0.988
2018	555	60.2	60.6	0.987
2019	562	65.4	65.3	0.985
2020	565	49.7	50.6	0.981
2021	575	72.7	72.6	0.992
2022	581	58.1	58.4	0.989
2023	586	63.9	64.3	0.985
2024	591	68.6	68.9	0.989
2025	595	69.4	69.9	0.986
All	6,232	62.3	62.6	0.989

Notes: Both decompositions are estimated on the same daily panel. *Aggregated* is the firm-specific information share from the two-variable VAR in $(r_{m,t}, r_{i,t})$; *Private + Public* is the sum of the private and public firm-specific shares from the three-variable VAR in $(r_{m,t}, x_{i,t}, r_{i,t})$. *Corr.* is the cross-sectional correlation between the two measures across stocks within the year. The bottom row pools all stock-years. The same pattern holds at quarterly frequency (correlation 0.90, mean absolute difference $\approx$ 6.9 pp across 24,810 stock-quarter cells).

As anticipated, the second and more consequential choice is to estimate the VAR at quarterly rather than annual frequency. Each firm contributes up to 44 observations to the decomposition instead of the 11 a yearly design would yield over the 2015–2025 window, and with 596 firms the quarterly design produces around 24,000 stock-quarter cells against at most $\sim$ 6,600 stock-year cells, enough to absorb both firm and time fixed effects without exhausting the degrees of freedom. A quarter is also the natural fundamental frequency of most of the independent variables, so aligning the decomposition window with the reporting window sharpens the interpretation of the subsequent regression of variance shares on ESG variables. The one trade-off is that each quarter contains only ≈ 65 trading days rather than ≈ 252, so every VAR is fit on a shorter time-series, but with a discrete gain in statistical power at the panel stage.

3.3 Panel regression specification

3.3.1 Estimating equation

The output of the variance decomposition is a firm-quarter panel with three columns: $\text{MktInfoShare}_{i,t}$, $\text{FirmInfoShare}_{i,t}$ and $\text{NoiseShare}_{i,t}$ in percentage points. The baseline specification is a two-way fixed-effects panel regression

$$Y_{i,t} \;=\; \alpha \;+\; \beta\, Z_{i,t} \;+\; \gamma' X_{i,t} \;+\; \mu_i \;+\; \lambda_t \;+\; \varepsilon_{i,t}, \tag{3.7}$$

where $Y_{i,t}$ is one of the three variance shares, $Z_{i,t}$ is the focal ESG-related variable, $X_{i,t}$ collects firm-level controls, μ_i and λ_t are firm and time fixed effects, and $\varepsilon_{i,t}$ is the residual. Three separate regressions are estimated, one per outcome, so that the sign and magnitude of β can be read off for each component of return variance.

3.3.2 Choice of variables and model specification

The Bloomberg ESG score is a slow-moving variable that is highly persistent within firm. The baseline therefore uses the quarter-on-quarter first difference, $\Delta\text{ESG}_{i,t} = \text{ESG}_{i,t} - \text{ESG}_{i,t-1}$, which isolates the news content of the score and is naturally aligned with a permanent-information story (a change in the rating should affect the price-formation environment going forward, not the level of past ratings).

The baseline control vector $X_{i,t}$ comprises two variables: log market capitalisation $\log \text{MktCap}_{i,t}$ and the quarterly realised volatility $\widehat{\sigma}_{i,t}$. The motivation is mechanical and conservative. Variance shares can be functions of firm size through the cross-section of liquidity, analyst coverage and institutional ownership; controlling for $\log \text{MktCap}$ neutralises the size channel and forces β to be identified from within-size variation in the ESG news content. Realised volatility enters because the variance shares are by definition functions of total return variance: a stock whose return variance is unusually high in quarter t will tend to have a mechanically different decomposition than the same stock in a quieter quarter, even if its information environment has not changed.

The baseline specification includes both entity (firm) and time fixed effects. Firm effects μ_i absorb time-invariant firm characteristics that simultaneously affect the information environment and the ESG profile of a firm: country of listing, industry, governance regime and ownership structure all fall under this heading. Time effects λ_t absorb euro-area macro-financial conditions, common volatility regimes (notably the covid-19 quarter 2020-Q1 and the energy-shock quarter 2022-Q1), and the EU-wide regulatory environment.

Inference uses standard errors clustered at the firm level, following the two-way fixed-effects literature (e.g., Bertrand et al., 2004). Within-firm serial correlation in variance shares is sizeable and clustering at the firm level provides asymptotically valid inference under arbitrary within-firm dependence.

A further complication is that neither $Y_{i,t}$ nor all of $X_{i,t}$ are directly observed. The three variance shares are themselves the output of a first-stage firm-quarter VAR, so each $Y_{i,t}$ carries first-stage sampling error that the second-stage regression cannot see; the realised-volatility control $\widehat{\sigma}_{i,t}$ is a generated regressor in the same sense. This is the classic *generated regressors* problem (Pagan, 1984): treating an estimated quantity as observed data in a subsequent regression ignores the first-stage estimation uncertainty and generally understates the standard errors of the second-stage coefficients (Murphy and Topel, 1985). The concern is not purely one of efficiency: if the precision of the firm-quarter decomposition is itself correlated with firm size or volatility, both of which are also regressors here, the induced measurement error in $Y_{i,t}$ is non-classical and could bias β rather than merely inflate its standard error. Clustering at the firm level absorbs part of this, since first-stage estimation noise for a given firm is serially correlated across quarters, but clustering does not correct for the fact that $Y_{i,t}$ and $\widehat{\sigma}_{i,t}$ are estimated rather than observed: it reduces, but does not eliminate, the understatement of uncertainty that the generated-regressors problem implies.

4 Results

This section reports the estimates of the two-way fixed-effects panel in equation (3.7). The variance decomposition delivers 24,810 firm-quarter cells for 595 stocks; taking the quarter-on-quarter first difference of the ESG score removes the first quarter of each firm and the handful of firms without two consecutive ESG observations, and listwise deletion on the controls trims a few further cells. The estimating sample is therefore a firm-quarter panel of 589 STOXX Europe 600 constituents over 43 quarters, for a total of 21,266 firm-quarter observations.

4.1 Baseline Estimates

Figure 4.1 plots the estimated coefficients on the change in the ESG score, $\Delta\text{ESG}_{i,t}$, on log market capitalisation, $\log \text{MktCap}_{i,t}$ and on quarterly realised volatility, $\widehat{\sigma}_{i,t}$, together with their 95% confidence intervals, for each of the three outcomes. Table 4.1 reports the same estimates in numerical form. As anticipated in Section 3.3, the control vector is deliberately minimal: the firm-level fundamentals available on the Bloomberg quarterly sheet were considered but contributed no explanatory power once firm and time fixed effects are in place.

The three shares sum to 100 by construction, so for any regressor the coefficients across the three equations sum to (approximately) zero: any variable that raises one share must lower the others by the same total amount. The coefficients on $\log \text{MktCap}_{i,t}$ are -1.515, 1.312 and 0.203, which sum to zero; those on $\widehat{\sigma}_{i,t}$ are -17.376, 18.062 and

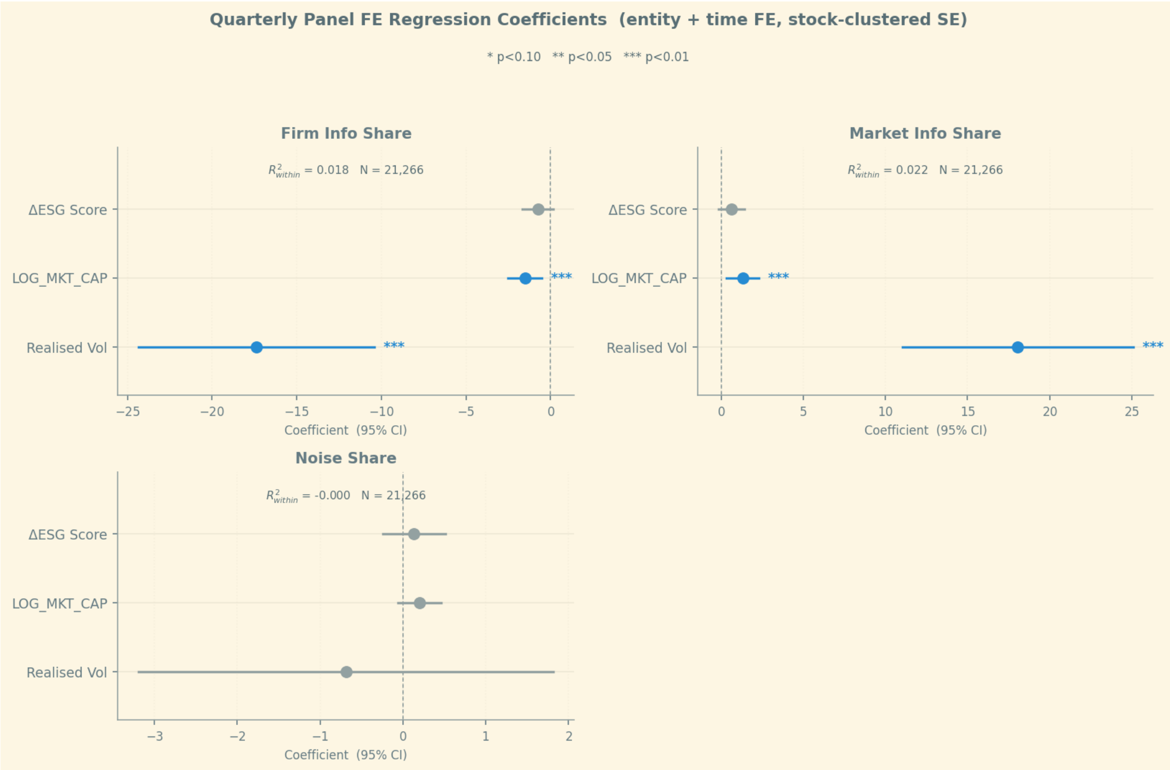

Figure 4.1: Quarterly panel fixed-effects coefficients

Notes: Each panel reports the coefficients of a two-way (firm and time) fixed-effects regression of one variance share (in percentage points) on the change in the Bloomberg ESG score, log market capitalisation and quarterly realised volatility $\widehat{\sigma}_{i,t}$ (in decimal units). Horizontal bars are 95% confidence intervals from standard errors clustered at the firm level; blue markers denote coefficients significant at the 10% level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4.1: Two-way fixed-effects estimates by variance share

	FirmInfoShare	MktInfoShare	NoiseShare
$\Delta \mathrm{ESG}_{i,t}$	-0.750	0.616	0.135
	(0.467)	(0.408)	(0.192)
$\log \mathrm{MktCap}_{i,t}$	-1.515^{***}	1.312^{***}	0.203
	(0.503)	(0.499)	(0.132)
$\widehat{\sigma}_{i,t}$	-17.376^{***}	18.062^{***}	-0.686
	(3.548)	(3.577)	(1.278)
Firm FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Observations	21,266	21,266	21,266
R^2 (within)	0.018	0.022	≈ 0.000

Notes: Outcomes are the firm-specific, market-wide and noise shares of return variance (percentage points). Firm-clustered standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

−0.686, again summing to zero. The regressions therefore describe how variance is *reallocated* across the information/noise components.

The first observation that can be done is that size and volatility trade off firm-specific against market-wide information, leaving noise untouched. Both controls behave exactly as the conservative, mechanical motivation in Section 3.3 would predict, and both load on the same margin. Larger firms have a significantly lower firm-specific information share and a correspondingly higher market-wide share (−1.515 and +1.312, both significant at the 1% level), consistent with the long-standing observation that the returns of large, widely-held stocks are driven disproportionately by systematic news and co-move more strongly with the market (Morck et al., 2000; Roll, 1988). The effect of realised volatility is the same in sign and an order of magnitude larger: a 0.10 increase in quarterly realised volatility lowers the firm-specific share by about 1.7 percentage points and raises the market-wide share by about 1.8, both highly significant. High-volatility quarters are thus predominantly market-wide episodes in which idiosyncratic variation is crowded out by systematic variation, the cross-sectional analogue of the rise in return synchronicity documented during turbulent periods (Rosch et al., 2017). Crucially, neither control has any detectable effect on the noise share: the coefficients are economically negligible and statistically indistinguishable from zero, and the within-R^2 of the noise regression is essentially zero. After firm and time fixed effects absorb the largely time-invariant, liquidity- and microstructure-driven cross-section of noise emphasised by Brogaard et al. (2022), what remains of the noise share has no systematic within-firm relationship to size or volatility. The controls move variance strictly along the firm-specific / market-wide axis.

Changes in the ESG score do not shift the composition of return variance. Regarding the ESG score coefficient, in none of the three equations is the estimate statistically distinguishable from zero at conventional levels: a one-point quarterly improvement in the ESG score is associated with a 0.75 percentage-point fall in the firm-specific information share ($p = 0.11$), a 0.62 point rise in the market-wide share ($p = 0.13$) and a near-zero change in noise ($p = 0.48$). The confidence intervals comfortably contain zero in every case. Two features of this result deserve emphasis. First, the noise channel, the one where disagreement across divergent ESG ratings (Berg et al., 2022) or attention-driven mispricing (Dim et al., 2023) would most plausibly show up, is the most precisely estimated null of the three: the point estimate is 0.135 with a standard error of 0.192, ruling out any but the smallest effects. Second, the sign of the (insignificant) point estimate on the firm-specific share is, if anything, the opposite of the prior that a higher ESG score should make prices incorporate more firm-specific information. Taken together, *the evidence does not support the hypothesis that within-firm changes in the ESG score reshape the news / noise composition of European equity-return variance.*

The synchronicity model of Section 2 gives these three coefficients a precise interpretation rather than leaving them as three unrelated numbers. A rise in κ_i (ESG-score revisions proxying the arrival of value-relevant firm information) should raise FirmInfoShare and lower the other two shares in equation (2.4); instead the point estimate on FirmInfoShare is negative. A taste-driven shock of the kind modelled by Pástor et al. (2022), entering through the systematic block $\kappa_m \beta_i f_t$ of equation (2.2), should raise MktInfoShare; the point estimate does have the predicted sign (+0.616) but is not significant. A rise in $\sigma^2_{u,i}$ from disagreement across divergent ESG ratings (Berg et al., 2022) should raise NoiseShare; instead this is the most precisely estimated zero of the three. None of the three channels the model isolates is detectably active, and, crucially, they are isolated separately: under the classical R^2 approach of equation (2.5), the information channel ($\kappa_i \uparrow$) and the disagreement channel ($\sigma^2_{u,i} \uparrow$) are observationally equivalent by equation (2.7), so a single synchronicity regression could not have told them apart even had it found a significant effect.

A null result on the focal variable is, in this design, informative rather than uninformative, and more informative than a null on $1-R^2$ would have been, for exactly the reason above: the three-share decomposition rules out the information channel and the disagreement channel independently rather than jointly. Beyond this identification advantage, three considerations counsel some caution before reading the null as a clean rejection of all three channels. First, the Bloomberg ESG score is a slow-moving composite rating, highly persistent and refreshed at low frequency (typically once or twice a year); its quarterly first-difference is zero or near-zero in most quarters, so $\Delta\text{ESG}_{i,t}$ carries little high-frequency signal. Second, the two-way fixed-effects design is deliberately demanding: firm effects remove the cross-sectional contrast between firms that are persistently ESG-strong and firms that are persistently ESG-weak, which is where much of the economically interesting variation is likely to lie, while time effects absorb precisely the EU-wide regulatory timing (the SFDR, Taxonomy and MiFID II milestones discussed in Section 3.1) through which ESG regulation acted as a common shock. What identifies β is the residual, idiosyncratic timing of firm-level score changes, a thin basis on which to detect a composition effect. Third, the channel emphasised by the theory may run through investor *attention* and *taste* rather than through the rating level, in which case the score is an imperfect proxy. These observations point to natural directions for sharpening the test, none of which the present firm-quarter design can deliver: a narrative or text-based measure of ESG attention in the spirit of Hassan et al. (2019), event-style identification around discrete rating or controversy revisions or a research design that exploits the cross-sectional rather than the within-firm variation in ESG exposure.

Set against the literature that motivated the research question in Section 1, the results are a mix of alignment and disagreement, and the pattern is itself informative. They align well with the measurement literature: if ESG ratings are as noisy and provider-dependent as Berg et al. (2022) document, and firm-level ESG disclosure builds up its informational infrastructure only slowly (Christensen et al., 2021), then a regression of variance shares on the quarterly change in a single vendor's score should be exactly this attenuated: the null is what that literature would predict from this specific proxy, not a surprise. The results also sit comfortably with the narrative-attention literature (Dim et al., 2023; Hassan et al., 2019): if price-relevant ESG information arrives through news flow and investor attention rather than through quarterly rating updates, a backward-looking, infrequently-refreshed rating is the wrong instrument for picking up the channel, and finding nothing with it says little about whether it exists.

Where the results sit less comfortably is against Pástor et al. (2022) directly. Their model and evidence say that shifts in aggregate green preferences move prices and could plausibly show up here as a reallocation toward MktInfoShare, which is the direction the point estimate points at. But that mechanism is, by construction, an aggregate one: an economy-wide or sector-wide taste shock hits many ESG-exposed firms in the same quarter, which is precisely the kind of variation the time fixed effects λ_t in equation (3.7) are designed to absorb, just as any persistent cross-sectional difference between ESG-strong and ESG-weak firms is precisely what the firm fixed effects μ_i absorb. What identifies β here is neither of those; it is the residual, idiosyncratic timing of one firm's rating update relative to its own history and to the quarter's common shock. Pástor et al. (2022)'s mechanism was never built to operate at that margin, so failing to detect it there is not evidence against their model; it is evidence that this design tests a different, more granular hypothesis than the one their aggregate taste-shock channel implies.

5 Conclusion

The research question asked in this work is whether the ascent of ESG investing has reshaped the composition of European equity-return variance as measured by the balance between firm-specific information, market-wide information and noise. To answer the question, the annual variance decomposition of Brogaard et al. (2022) was adapted to a quarterly frequency and a European cross-section through the deviations documented in Section 3.2, producing firm-quarter variance shares for 589 STOXX Europe 600 constituents over 2015–2025. These shares were then related to within-firm changes in the Bloomberg ESG score in a double firm-time fixed-effects panel.

Two findings stand out. First, within-firm changes in the ESG score have no detectable effect on any of the three variance shares: none of the information, taste-shock or disagreement channels formalised in Section 2 is detectably active, and the noise coefficient (where disagreement across divergent ESG ratings (Berg et al., 2022) would show up) is the most precisely estimated null of the three. Second, the two mechanical controls, firm size and realised volatility, reallocate variance cleanly along the firm-specific / market-wide axis and leave the noise share untouched, with high-volatility quarters tilting variance towards the systematic component, the cross-sectional counterpart of the rise in synchronicity observed in turbulent periods.

The contribution is twofold. Methodologically, the paper shows that the Brogaard decomposition can be taken to quarterly frequency without losing its interpretability, extracting substantially more information from the same daily price panel than an annual design would (see Section 3.2.2). Moreover, the comparison in Table 3.5 confirms that dropping the volume equation costs almost nothing and avoids making some specific assumptions about the market micro-structure. Substantively, the null is itself informative: the synchronicity model of Section 2 shows that a rise in firm-specific information and a rise in noise push $1 - R^2$ in the same direction (equation (2.7)), so the classical R^2-based literature could not have separated them (Morck et al., 2000; Roll, 1988) even had it detected an effect; the three-share decomposition rules out both channels independently and finds that neither responds to the ESG score, at least as measured by a slow-moving rating at the firm-quarter level. This is not evidence against the taste-shock mechanism of Pástor et al. (2022): that mechanism is an aggregate one, and the firm and time fixed effects here are built to absorb exactly the aggregate and persistent cross-sectional variation through which it would operate, so the design speaks to a narrower, within-firm question than their model addresses.

These conclusions must be read against the limitations of the measure. The original design was built not around the Bloomberg ESG score but around a *narrative* index: a text-extracted measure of firm-level ESG attention, something similar to what Hassan et al. (2019) did, intended to capture the arrival of ESG-relevant news rather than the level of a rating. That measure had to be abandoned for lack of usable data at the firm-quarter frequency and cross-sectional coverage the design required, and the ESG score was adopted as a fallback. This matters for interpretation: the score is highly persistent and refreshed only once or twice a year, so its quarterly first-difference is zero in most quarters and the test is run on a thin and possibly mis-specified signal, while the two-way fixed effects additionally absorb both the cross-sectional contrast between ESG-weak and ESG-strong firms and the EU-wide regulatory timing. Reconstructing the narrative index, together with event-style identification around discrete rating or controversy revisions and a cross-sectional design that exploits the between-firm variation the fixed effects here remove, is the natural next step and the most promising route to a sharper test of whether the ESG regulatory trend reshapes the information content of prices.

A second limitation lies in the econometric specification of the decomposition itself, inherited directly from Brogaard et al. (2022): the VAR fitted in each stock-quarter is a constant-coefficient, constant-variance system, so both the loadings θ and the shock variances σ_ε^2 are implicitly treated as fixed within the window and left to jump discontinuously across quarter boundaries. Neither assumption sits comfortably with well-documented features of daily returns. Volatility clustering means σ_ε^2 almost certainly varies within a quarter, and the market-wide-share spikes around the 2020-Q1 and 2022-Q1 shocks visible in Figure 3.1 are themselves evidence of within-sample regime shifts that a constant-parameter model can only average over rather than track. A time-varying-parameter VAR with stochastic volatility (Primiceri, 2005) would let θ_t and $\sigma_{\varepsilon,t}^2$ evolve smoothly instead of resetting every quarter and, as a state-space model, would deliver filtered uncertainty bands for θ_t that could be propagated into the second-stage panel regression, directly addressing the generated-regressors concern raised in Section 3.3. This is a more demanding estimation problem, but a possible and tractable next step.

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